

$$d_{it} = (1 - \kappa) \left( \sum_{k=1}^K \kappa^{k-1} r_{t+k,1t} - \sum_{k=1}^K \kappa^{k-1} \Delta d_{t+k,1t} \right) + \kappa^K \cdot d_{t+K,1t} \quad (1.7)$$

$$= (1 - \kappa) \left( \sum_{k=1}^K \kappa^{k-1} R_t[r_{t+k,1t}] - \sum_{k=1}^K \kappa^{k-1} R_t[\Delta d_{t+k,1t}] \right) + \kappa^K \cdot R_t[d_{t+K,1t}] \quad (1.8)$$

$$= (1 - \kappa) \left( \sum_{k=1}^K \kappa^{k-1} R_t[r_{t+k,1t}] - \sum_{k=1}^K \kappa^{k-1} R_t[\Delta d_{t+k,1t}] \right) \quad (1.9)$$

where  $d_{it} = \log(1 + D_t/P_t)$  and  $\kappa = e^{-\bar{r}_t}$ .

Starting from  $R_{t,t+1} = (P_{t+1} + D_{t+1})/P_t$ , derive the (approximate) Equations 1.7, 1.8, and 1.9. Explain and justify any assumptions you make.<sup>5</sup>

First, assume  $D > 0$

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}$$

$$\begin{aligned} R_{e,t+1} \times \frac{P_t}{R_{t-1}} &= \frac{P_{t+1} + D_{t+1}}{P_t} \times \frac{P_t}{R_{t-1}} \\ &= \frac{P_t}{P_t} \times \frac{P_{t+1} + D_{t+1}}{R_{t-1}} \\ &= \frac{R_t + D_t}{P_t} \times \frac{P_t}{P_t + D_t} \\ &= \frac{R_{t-1}}{P_{t+1} + D_{t+1}} \end{aligned}$$

So we have

$$R_{e,t+1} \times \frac{P_t}{R_{t-1}} = \frac{\frac{R_t + D_t}{P_t} \times \frac{P_t}{P_t + D_t}}{\frac{R_{t-1}}{P_{t+1} + D_{t+1}}}$$

Take log on both sides:

$$\log(R_{e,t+1} \times \frac{P_t}{R_{t-1}}) = \log\left(\frac{\frac{R_t + D_t}{P_t} \times \frac{P_t}{P_t + D_t}}{\frac{R_{t-1}}{P_{t+1} + D_{t+1}}}\right)$$

$$\log(R_{e,t+1}) + \log\left(\frac{P_t}{R_{t-1}}\right) = \log\left(\frac{R_t + D_t}{P_t}\right) + \log\left(\frac{P_t}{P_t + D_t}\right) - \log\left(\frac{R_{t-1}}{P_{t+1} + D_{t+1}}\right)$$

$$\text{def } Y_{e,t+1} = \log(R_{e,t+1}) \quad P_t = \log(P_t) \quad d_t = \log(D_t) \quad dy_t = \log\left(1 + \frac{D_t}{P_t}\right)$$

$$\begin{aligned} \text{given that } \log\left(\frac{P_t}{P_t + D_t}\right) &= \log\left(1 - \frac{D_t}{P_t + D_t}\right) \\ &= \log\left(1 - e^{\log\left(\frac{-D_t}{P_t + D_t}\right)}\right) \\ &= \log\left(1 - e^{-\log\left(\frac{D_t + P_t}{D_t}\right)}\right) \\ &= \log\left(1 - e^{-\log\left(1 + \frac{P_t}{D_t}\right)}\right) \\ &= \log\left(1 - e^{-dy_t}\right) \end{aligned}$$

$$\text{Thus, } \log\left(\frac{P_t}{P_t + D_t}\right) = \log\left(1 - e^{-dy_t}\right)$$

$$\text{Similarly } \log\left(\frac{P_{t+1}}{P_{t+1} + D_{t+1}}\right) = \log\left(1 - e^{-dy_{t+1}}\right)$$

We have

$$Y_{e,t+1} - \Delta d_{t+1} = dy_t + \log(1 - e^{-dy_t}) - \log(1 - e^{-dy_{t+1}})$$

$$f(x) = \log(1 - e^{-x})$$

$$f'(x) = \frac{1}{1 - e^{-x}} \cdot e^{-x} = \frac{e^{-x}}{1 - e^{-x}}$$

let  $x = dy_t$ , by first-order approximation, def  $k \equiv e^{-\bar{dy}}$

$$\begin{aligned} \log(1 - e^{-dy_t}) &\approx \log(1 - e^{-\bar{dy}}) + \log'(1 - e^{-\bar{dy}})(dy_t - \bar{dy}) \\ &= \log(1 - k) + \frac{e^{-\bar{dy}}}{1 - e^{-\bar{dy}}}(dy_t - \bar{dy}) \\ &= \log(1 - k) + \frac{k}{1 - k}(dy_t - \bar{dy}) \end{aligned}$$

Thus, we have

$$\text{Take } \log(1 - e^{-dy_t}) \approx \log(1 - k) + \frac{k}{1 - k}(dy_t - \bar{dy})$$

$$\text{similarly } \log(1 - e^{-dy_{t+1}}) \approx \log(1 - k) + \frac{k}{1 - k}(dy_{t+1} - \bar{dy})$$

$$\text{inside } Y_{e,t+1} - \Delta d_{t+1} = dy_t + \log(1 - e^{-dy_t}) - \log(1 - e^{-dy_{t+1}}),$$

We have

$$\begin{aligned}
 Y_{e,t+1} - \Delta d_{t+1} &= dy_t + \log(1-k) + \frac{k}{1-k} (dy_t - \bar{dy}) - \log(1-k) - \frac{k}{1-k} (dy_{t+1} - \bar{dy}) \\
 &= dy_t + \frac{k}{1-k} (dy_t - dy_{t+1}) \\
 &= \frac{1}{1-k} dy_t - \frac{k}{1-k} dy_{t+1} \\
 \frac{1}{1-k} dy_t &= Y_{e,t+1} - \Delta d_{t+1} + \frac{k}{1-k} dy_{t+1} \\
 dy_t &= (1-k)(Y_{e,t+1} - \Delta d_{t+1}) + k \cdot dy_{t+1}
 \end{aligned}$$

$$\begin{aligned}
 dy_t &= (1-k)(Y_{e,t+1} - \Delta d_{t+1}) + k \cdot dy_{t+1} \\
 &= (1-k)(Y_{e,t+1} - \Delta d_{t+1}) + k [(1-k)(Y_{e,t+2} - \Delta d_{t+2}) + k \cdot dy_{t+2}] \\
 &\quad \vdots \text{ keep substituting } dy_{t+h} \\
 &= (1-k) \left( \sum_{h=1}^H k^{h-1} Y_{e,t+h} - \sum_{h=1}^H k^{h-1} \Delta d_{t+h} \right) + k^H dy_{t+H}
 \end{aligned}$$

Thus, we have

$$dy_t = (1-k) \left( \sum_{h=1}^H k^{h-1} Y_{e,t+h} - \sum_{h=1}^H k^{h-1} \Delta d_{t+h} \right) + k^H dy_{t+H}$$

Take expectation at  $t$ , we have

$$dy_t = (1-k) \left( \sum_{h=1}^H k^{h-1} E_t[Y_{e,t+h}] - \sum_{h=1}^H k^{h-1} E_t[\Delta d_{t+h}] \right) + k^H E_t[dy_{t+H}]$$

Suppose  $H \rightarrow \infty$ , given that  $k = e^{-\bar{dy}} \in (0, 1)$ ,  $\lim_{H \rightarrow \infty} k^H = 0$

And because of the no-bubble condition,  $\lim_{H \rightarrow \infty} k^H E_t[dy_{t+H}] = 0$

we have

$$dy_t = (1-k) \left( \sum_{h=1}^H k^{h-1} E_t[Y_{e,t+h}] - \sum_{h=1}^H k^{h-1} E_t[\Delta d_{t+h}] \right)$$